MengerGL – 3D graphics Programming

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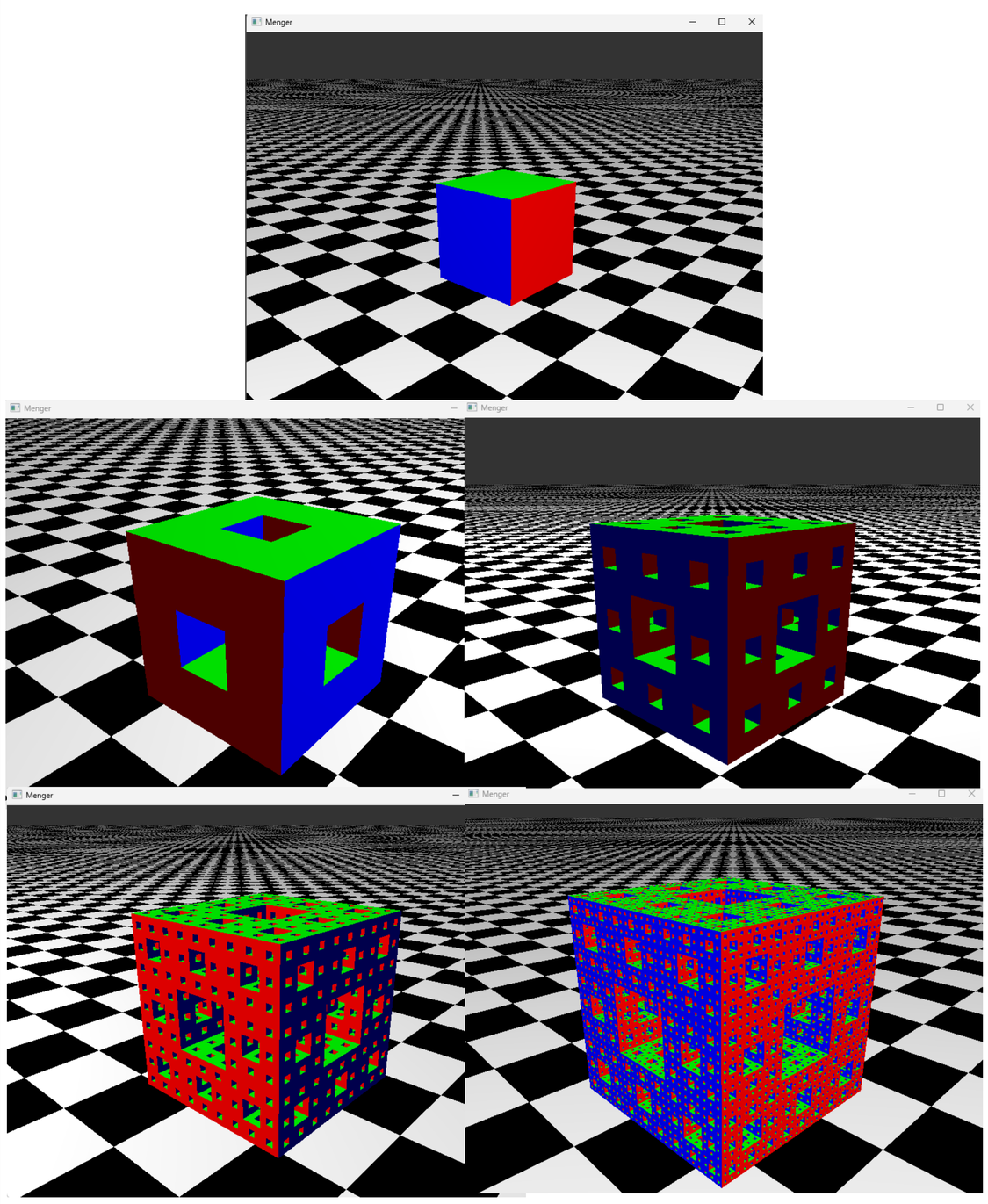
Introduction

MengerGL is a real-time 3D rendering project that visualizes the Menger Sponge fractal using OpenGL. It uses recursive subdivision and GPU acceleration to render the structure efficiently. The sponge has the unique property of increasing surface area while its volume approaches zero. The project demonstrates key graphics concepts like geometry generation, shaders, and camera controls.

Visualizations



- Fractal 1 – Base Cube no subdivisions
- Fractal 2 (top left) – 6 total subdivisions (1 on each face)
- Fractal 3 (top right) – 20 total subdivisions (corners and edges remain)
- Fractal 4 (bottom left) – 400 total subdivisions (each cube subdivided further)
- Fractal 5 (bottom right) – 8000 total subdivisions (increasing fractal complexity)

Implementation

- Geometry is stored in vertex and normal buffers
- This removal process creates tunnels through the center of each face and the middle of the structure
- Face indices are generated by grouping every three vertices into triangles
- OpenGL uses these triangles to render the final mesh
- Each iteration level uses a dedicated helper function that recursively calls the previous level

Results and Performance

Levels	Vertices	Triangles	FPS
1	720	240	~6,000
2	14,400	4,800	~4,800
3	288,000	96,000	1,000+
4	5,760,000	1,920,00	~60

- The recursive architecture is both accurate and computationally efficient
- Each iteration correctly matches the mathematical definition of the Menger Sponge
- The isTunnel() function consistently removes the correct 7 cubes at every level
- The program maintains around 60 FPS even at high complexity
- Real-time performance is achieved without advanced optimizations

Acknowledgement

Sincere thanks to Dr.Tang for his support and instruction on this project!